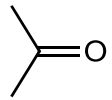
3(a)(i)		**1**
3(b)(i)	$(CH_3)_2CHOH + Na \rightarrow (CH_3)_2CHO^{(-)}Na^{(+)} + \frac{1}{2}H_2$	**1**
3(b)(ii)	**M1** correct dipole on $^{\delta+}C—Br^{\delta-}$ **AND** curly arrow from C—Br bond to Br **M2** intermediate  **M3** curly arrow from lone pair of $(CH_3)_2CHO^-$ to C^+	**3**
3(b)(iii)	**M1** rate would decrease **M2** C—Br is weaker (than C—C*l*, so breaks more easily)	**2**

Examiner report missing
9701_w25_qp_24 Question 3

4(a)(i)	**M1** (molecules with the) same molecular formula / same number of atoms of each element **M2** different structural formulae/structures	**2**

4(b)(i)	KCN in ethanol (and heat)	**1**

Examiner report missing
9701_w25_qp_24 Question 4

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	**1**

3(d)(i)	HC*l* (g)	**1**

Examiner report missing
9701_w25_qp_23 Question 3

4(a)(i)	reduction	**1**

4(a)(ii)	PCl_5 / PCl_3 / $SOCl_2$ **OR** KCl / NaCl **AND** conc. H_2SO_4 / conc.H_3PO_4	**1**

Examiner report missing
9701_w25_qp_23 Question 4

4(a)(i)	free-radical substitution	**1**
4(b)	sodium hydroxide **AND** water	**1**

Examiner report missing
9701_w25_qp_22 Question 4

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	**1**

3(d)(i)	HC*l* (g)	**1**

Examiner report missing
9701_w25_qp_21 Question 3

4(a)(i)	reduction	**1**

4(a)(ii)	PCl_5 / PCl_3 / $SOCl_2$ **OR** KCl / NaCl **AND** conc. H_2SO_4 / conc.H_3PO_4	**1**

Examiner report missing
9701_w25_qp_21 Question 4

4(a)	$C_5H_{10}O_2$	**1**

4(d)(iii)	**M1** step 1 PCl_3 and heat **OR** HCl(g) **OR** PCl_5 **OR** $SOCl_2$ **OR** $NaCl$ / KCl and conc. H_2SO_4 / conc. H_3PO_4 **M2** step 2 reagent KCN **M3** conditions for step 2 dissolved in ethanol **AND** heat	**3**

Examiner report missing
9701_s25_qp_24 Question 4

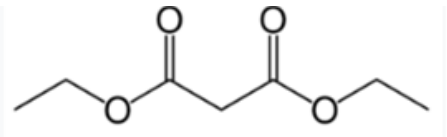
6(a)(i)	two or more reactants / molecules react to make (only) one product	**1**

6(c)(i)	KCN/potassium cyanide in ethanol **AND** heat	**1**

Examiner report missing
9701_s25_qp_23 Question 6

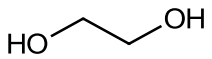
4(a)(i)	$LiAlH_4$	**1**
4(b)(i)	substitution	**1**
4(b)(ii)	KCN / potassium cyanide in ethanol / C_2H_5OH **AND** heat	**1**

Examiner report missing
9701_s25_qp_22 Question 4

5(a)	**M1** two ester links **M2** rest of structure correct 	**2**

5(d)(i)	**M1** NaOH(aq) **AND** heat **M2** acidified **AND** potassium dichromate(VI) **AND** heat under reflux	**2**

Examiner report missing
9701_s25_qp_21 Question 5

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	**2**
1(b)(i)	green colour (of gas) disappears	**1**
1(b)(iv)	phosphoric(V) acid	**1**
1(b)(v)		**1**
1(c)(i)	contains a(/ one or more) C=C / double / multiple bonds	**1**
1(c)(ii)	ethanol (solvent) **AND** heat	**1**

Examiner report missing
9701_m25_qp_22 Question 1

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**

2(b)(ii)	ultraviolet	**1**

2(c)(iv)	**M1 white** precipitate/**white** ppt forms	**1**
	M2 $CHCl_3$ is hydrolysed / chloride ions are released **OR** Cl^- / chloride ions which are released react with Ag^+ **OR** silver chloride / $AgCl$ is formed	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

(ii) Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

(iii) Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

(iv) Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

(b) (i) Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

(ii) This question was well answered.

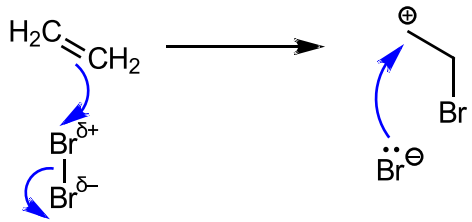
(iii) Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

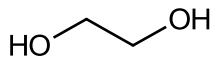
(c) (i) Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

(ii) The definition of disproportionation was well answered by many candidates.

(iii) This equation was well known.

(iv) This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of NO_2 is also formed. Those candidates who described how the Cl^- ion was formed, prior to its reaction with Ag^+ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form Cl^-. Many stated that the Cl^- was contained in $CHCl_3$, and not a product from its hydrolysis.

4(a)	 **pt 1** dipole on Br_2 **pt 2** curly arrow from C=C bond to a $Br^{\delta+}$ **pt 3** curly arrow from Br–Br bond to the other Br **pt 4** correct intermediate **pt 5** curly arrow from lone pair on Br^- to C^+ **5•** = 3 marks. **4•** = 2 marks. **3•** = 2 marks. **2•** = 1 marks. **1•** = 0 marks.	**3**

4(c)(i)		**1**
	bromoethene	**1**

4(c)(iii)	C_2H_2	**1**

Question 4

(a) Well-rehearsed candidate answers included the accurate placement of curly arrows, dipoles and the correct intermediate. Weaker answers often had ambiguous placement of curly arrows, inappropriate use of partial charges, incorrect dipole on Br_2 and incorrect intermediates.

(b) Correct calculations, sometimes with an appropriate Hess' cycle in their working, were seen. Incorrect answers were calculated by those candidates that did not fully understand or use the definition of enthalpy change of formation when applying Hess's law.

(c) (i) The product **A** was correctly shown in many answers. Candidates need to show correct connectivity of the O-H group to the carbon backbone when structures are drawn.
The naming of **B** presented problems for some candidates. Incorrect names often described the incorrect position of Br on the carbon backbone or gave a name which described the wrong number of carbon atoms in the backbone.

(ii) Unambiguous structures were seen in a significant number of candidate answers.

(iii) This question proved to be extremely demanding. All the information provided in the details of Fig. 5.1 needed to be considered, including the three products as well as the reaction conditions.

(d) (i) Correct identification of the type of isomerism was given in a significant number of candidate answers. Chain isomerism was a common incorrect answer even though the two molecules were made up of only two carbon atoms.

(ii) Many candidate answers stated that **F** was a carbonyl compound; this term describes the functional group present in the molecule and is common to both homologous series; aldehyde and ketone.

(iii) Many candidate answers gave appropriate observations for the behaviour of **F** with 2,4-DNPH reagent and Tollen's reagent. A much smaller number of answers identified a positive reaction with appropriate observation for the reaction with alkaline I_2(aq).

(e) (i) This question proved to be demanding. The strongest candidates gave full explanations of their interpretation of the infrared spectrum by relating relevant wavenumber values at absorption troughs to appropriate bond types and used the molecular ion peak to deduce the molecular mass of **H**. Weaker responses quoted key wavenumber values at troughs in the infrared spectrum and restated that the molecular ion peak in the mass spectrum was at m/e = 60 without explaining what this information actually provides about structure **H**.

(ii) The correct role of reagent **G** was given in many candidate answers. Catalyst was a common incorrect answer.

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**

2(b)(ii)	ultraviolet	**1**

2(c)(iv)	**M1 white** precipitate/**white** ppt forms	**1**
	M2 $CHCl_3$ is hydrolysed / chloride ions are released **OR** Cl^- / chloride ions which are released react with Ag^+ **OR** silver chloride / $AgCl$ is formed	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

(ii) Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

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(ii) This question was well answered.

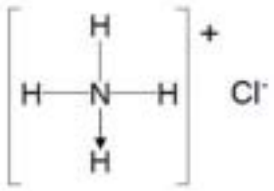
(iii) Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

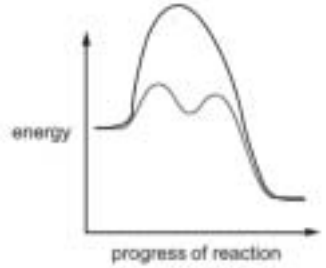
(c) (i) Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

(ii) The definition of disproportionation was well answered by many candidates.

(iii) This equation was well known.

(iv) This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of NO_2 is also formed. Those candidates who described how the Cl^- ion was formed, prior to its reaction with Ag^+ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form Cl^-. Many stated that the Cl^- was contained in $CHCl_3$, and not a product from its hydrolysis.

4(a)	 **OR** Correct representation or label showing **M1** (ionic bond between) NH_4^+ and Cl^- **AND** 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 **M2** 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons1	**2**
4(c)(i)	nucleophile	**1**
4(c)(ii)	OH^- (more reactive because) negatively charged species more attracted to halogenoalkane and / or $C^{\delta+}$ and / or intermediate and / or C^+ **OR** NH_3 (more reactive because) N electrons less tightly held **OR** N is more able to donate (lp) electrons (to make the covalent bond) **OR** N is less electronegative ORA	**1**

4(d)	 **M1** same start energy and finishing energy **AND** single hump **M2** transition state / activation energy shown in answer is higher than both transition state energies shown in reaction profile (as the reaction is slower)	**2**
4(e)(i)	**M1** dipole and correct arrow on C—Br of 2-bromobutane **M2** lone pair on NH_3 attacking C(+) of the intermediate drawn **M3** curly arrow showing bond between N—H breaking and pointing towards the N	**3**
4(e)(ii)	NH_4Br / ammonium bromide	**1**
4(f)(i)		**1**
4(f)(ii)	**M1** 2-bromo-2-methylpropane **AND** more stable carbocation / C^+ **M2** alkyl groups have an inductive effect (of alkyl groups) **AND** 2-bromo-2-methylpropane has more / most alkyl groups attached (to C^+) / carbocation contains more alkyl groups	**2**

Question 4

Many individual parts of this question were answered well. Parts **(a)**, **(c)(i)**, **(e)(ii)** and **(f)(ii)** were clearly the most challenging, often as candidates were required to apply several independent parts of their knowledge of the syllabus.

(a) Candidates were not strictly required to give dot-and-cross diagrams, though these and modified Lewis diagrams often produced the clearest responses. Many answers incorrectly showed a pentavalent nitrogen atom — a kind of NH_5 species, with little understanding of the NH_4^+ Cl^- superstructure. The dative bond needed to be clearly distinct from other covalent bonds and ionic charges needed to be given.

(b)(i) There were many correct answers to this question.

(ii) Different versions of this answer were offered, but a large number were a simple addition of NH_4^+ and OH^-, which did not gain credit.

(c)(i) This was well answered by candidates whose knowledge extended to terminology associated with electron-pair donors.

(ii) This was found to be the most challenging item on the paper which required candidates to assess either the basicity or nucleophilicity of each species. Opposing answers were therefore acceptable with enough reasoning.

(d) This was generally well answered by candidates, but the single-step nature of the S_N2 mechanism needed to be clearly evidenced by one transition state; the slower rate of this reaction by a higher energy of the TS relative to either produced in S_N1. Candidates frequently accessed the former part but often then erroneously showed some kind of catalysed reaction with a lower activation energy.

(e)(i) This question was not generally well answered, as candidates were not able to associate electron movement with the correct stage of the reaction and often were very imprecise with their positioning of curly arrows. Many answers had the arrowhead and tail reversed, with a lot of responses suggesting electron movement *from* extra H^+ species.

(ii) This question was reasonably well answered.

(iii) This question was generally well answered, although stray vowels or unspecific/incorrect numbering were observed. Candidates are reminded that the longest carbon chain is the important root of a molecule's name.

(f) (i) Candidates were able to provide a correct answer here by analogy of the 2-bromobutylium species on the left-hand side.

(ii) This was found to be a demanding question, although candidates who recognised the necessary reference to the stability of the intermediate were able to give a reasonably straightforward AO1 account for full credit.

5(a)	cold dilute $KMnO_4$ **OR** cold dilute potassium manganate(VII)	**1**
5(b)(i)	step 1 …..Br_2 (in the dark) step 2 …..NaOH(aq) + heat	**2**
5(d)(i)	**W** =	**1**

Question 5

This question tested organic reactions and spectroscopic methods using cyclohexene as a starting material. Many candidates seemed familiar with the chemistry, although the context of a cyclic molecule proved to be disconcerting for a few, who appeared to misinterpret the information given, or did not concentrate solely on the relevant functional groups.

(a) This was well answered by many, although differences appeared in the conditions to use with the $KMnO_4$ reagent (e.g. concentrated/dilute, alkaline/neutral/acidic).

(b)(i) Step 1 was familiar to many candidates. The hydrolysis of two functional groups in step 2 led many to think that H_2O_2 was being used, rather than two portions of NaOH.

(ii) In asking for a mechanism, 'electrophilic' was expected as well as 'addition'.

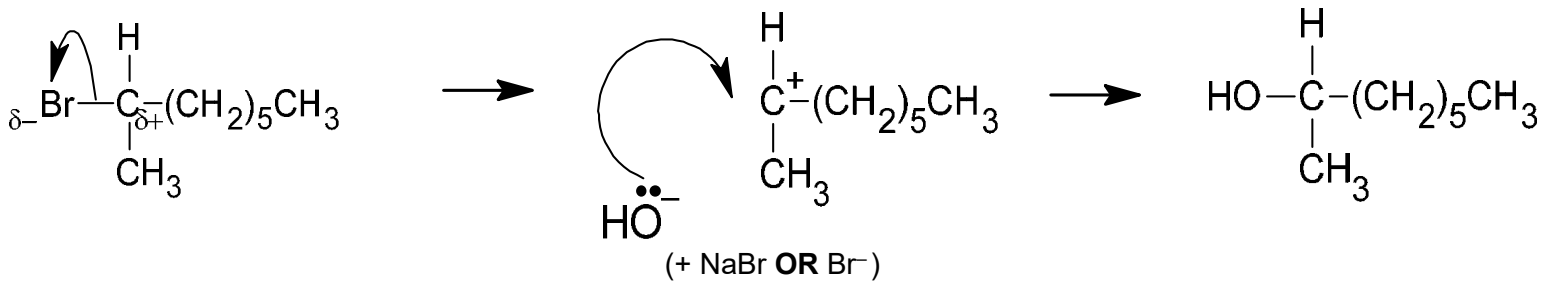
(c) Candidates showed some understanding in answering this question but often missed full credit by not following the rubric properly: the diagram needed labelling and each responsible bond (stretch) needed to be clearly identified. Many candidates only achieved half of this, or switched **F** and **G**.

(d)(i) This was mostly well answered, although some candidates lost credit when opting to give fully displayed versions of the CN group, neglecting to show a triple C≡N bond or using the wrong connectivity (N attached to the ring).

(ii) This was answered well by many candidates. Others did not acknowledge that acidic conditions are used, creating an ammonium species rather than NH_3.

(iii) The formulae of the organic products were generally well indicated, though candidates often then misbalanced the equation with insufficient (H).

(iv) This was generally well answered by candidates. A few $NaBH_4$ responses were offered, to no credit: carboxylic acids need the stronger of the two hydride reagents on the syllabus.

4(a)	Three-dimensional stereoisomers of $CH_3(CH_2)_5CHBrCH_3$ **M1** correct representation of left-hand answer as 3d structure of one of the enantiomers of $CH_3(CH_2)_5CHBrCH_3$ **M2** correct representation of 3d structure of the second enantiomer of $CH_3(CH_2)_5CHBrCH_3$	**2**
4(b)	 (+ NaBr **OR** Br^-) **M1** correct dipole ($^{\delta+}C—Br^{\delta-}$) **AND** curly arrow from bond of C—Br to Br (or just beyond) **M2** correct intermediate **ALLOW** R for $–(CH_2)_5CH_3$ **M3** curly arrow from lone pair on OH^- to the C of intermediate (or suitable position where C–OH bond is made) *If $SN2$ mechanism shown:* Apply $SN2$ mechanism mark scheme **only** if mechanism shows OH / NaOH species attacking the reactant and no OH **M1** correct dipole ($^{\delta+}C—Br^{\delta-}$) AND curly arrow from bond of C—Br to Br of reactant (or just beyond) **NOT M2** **M3** curly arrow from lone pair on $:OH^-$ to the appropriate C of the reactant	**3**

4(c)	<table><tr><td>Chemical test</td><td>observation with 2-bromooctane</td><td>observation with octan-2-ol</td><td>observation with $CH_3(CH_2)_5CHCH_2$</td></tr><tr><td>Br_2 in the dark</td><td>×</td><td></td><td>orange / brown to colourless</td></tr><tr><td>PCl_5</td><td></td><td>steamy fumes</td><td>×</td></tr><tr><td>$AgNO_3(aq)$</td><td>cream precipitate **OR** off white precipitate</td><td>×</td><td></td></tr></table> 6 correct answers = 3 marks 4 or 5 correct answers = 2 marks 2 or 3 correct answers = 1 mark 0 or 1 correct answer = 0 marks	**3**
4(d)(i)	elimination	**1**
4(d)(ii)	NaOH / sodium hydroxide in ethanol	**1**

Question 4

(a) Some excellent representations of the three-dimensional structures were seen; these included the approximate bond angles for these tetrahedral structures. Other diagrams showed confusion over the types of symbols used to represent bonds in the plane with those in and out of the plane. In order to remove ambiguity, candidates should be encouraged to use the convention described in the syllabus, under expected conventions for representing organic structures.

(b) Good knowledge of the S_N1 mechanism was described with precision in the best answers. Some confusion regarding the relative charges on the C–Br dipole and either no reference to a lone pair of electrons on O of OH^- or the location of the lone pair of electrons on the H of OH^- were common incorrect details. Some answers confused the S_N2 mechanism details with S_N1.

(c) The observations of these reactions were generally well known. Some answers did not include observations, as directed in the table heading, rather than attempted to predict the product of the reactions.

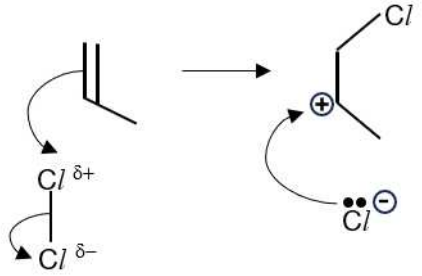
(d) (i) The name of this reaction was quite well known. Addition, substitution, dehydration and cracking were common incorrect answers.

(ii) The reagent NaOH was described in many answers. Confusion of the relevant conditions was seen in weaker responses.

(e) (i) Fully correct answers were seen in around half of responses. Sometimes confusion between two of the types of hybridisations was seen. The weakest answers demonstrated a lack of understanding of the different types of hybridisations which occur in carbon atoms in order to produce single, double and triple covalent bonds.

(ii) It was common for answers to contain features which lead to both types of stereoisomerism rather than focus on the stereoisomerism present in unbranched hydrocarbons. A relatively small proportion of answers showed good understanding of the features of an alkene required in order for it to exist as geometrical (cis/trans) isomers. A lack of precision in answers produced some ambiguity in responses, for example ‘no movement of a double bond’ rather than ‘no rotation of a double bond’.

2(a)	<table><tr><td>formula of chloride</td><td>● $NaCl$</td><td>● $MgCl_2$</td><td>● $SiCl_4$</td></tr><tr><td>structure</td><td>giant</td><td>● giant</td><td>● simple</td></tr><tr><td>bonding of chloride</td><td>● ionic</td><td>● ionic</td><td>covalent</td></tr><tr><td>pH of solution formed</td><td>● 7</td><td>6.2</td><td>● 1–4</td></tr></table>	**3**

2(d)(i)	 ● curly arrow from C=C bond to $Cl^{(\delta+)}$	**1**
	● correct dipole on Cl_2 **AND** curly arrow from Cl–Cl bond to $Cl^{\delta-}$	**1**
	● correct carbocation intermediate	**1**
	● curly arrow from lone pair on Cl^- to C^+	**1**

2(d)(ii)	NaOH in ethanol **AND** heat	**1**

Question 2

This was a holistic question based around the chemistry of chlorine and chlorides. Candidates are recommended to give brief answers to one-mark questions and to draw diagrams clearly.

(a) Common errors were found in the formulae of the chlorides, the structure of $SiCl_4$ and in the pH of a $NaCl$ solution (neutral so pH 7; candidates often seemed to think that: $NaCl + H_2O \rightarrow HCl + NaOH$).

(b) (i) Many variant and phonetic spellings of 'disproportionation' were observed.

(ii) There was an issue with this question in that water was not included as one of the products of the reaction, so full marks were awarded to all candidates to make sure no candidates were disadvantaged. This has been corrected in the published version of the paper.

(iii) This equation was rarely constructed correctly: where more valid attempts were made, common errors were either that $NaClO$ was given as the oxidation product or that the correct stoichiometric coefficients were omitted.

(iv) This was answered well by most candidates, although a number offered effervescence as an observation.

(c) (i) Octahedral was the most common answer for PCl_6^-, although commonly hexagonal or octagonal were seen (and not credited). A broad range of answers were given for PCl_4^+, with 'trigonal pyramidal' (missing the bi- prefix) and 'bipyramidal' (missing the base shape) among them.

(ii) **J** was often correctly identified as H_2O/water. Hydration or displacement were often given instead of hydrolysis or substitution for the type of reaction.

(d) (i) Some very loosely drawn diagrams were unable to be given full credit: curly arrows should come from a clear origin (the middle of a bond, or from pair of electrons, rather than a negative charge) and, in cases such as this, the arrowhead needs to have both 'blades' and point to a specific atom or the middle of a bond.

Many candidates also confused partial charges/dipoles with full formal charges.

(ii) Candidates were required to reference heating the ethanolic NaOH to gain full credit.

(iii) Many answers retained the C=C or arranged the 3C monomer linearly with no branching. These were both common and recurring errors in questions of this type.

4(a)	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	**1**
	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> $CH_3CHBrCH_2CH_3$	**1**
4(b)(i)	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr></table>	**1**
4(b)(ii)	<table><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> $CH_3CHBrCH_2CH_3$ nucleophilic substitution	**1**

Question 4

This an organic and spectroscopic question. Candidates are advised to be clear in matters of structure, both when representing molecules diagrammatically and when referring to features within a molecule, such as those identified by different types of spectra.

(a) This was answered well by many candidates. There were incidences of ambiguous answers, such as a 'crossed tick' (✓✗). Credit cannot be awarded if an answer is ambiguous.

(b) (i) Unambiguous structural formulae only were accepted, therefore molecular formulae were discounted and those in which the connectivity was questionable.

(ii) There were many correct answers to this question.

(iii) This proved to be one of the most challenging questions on the paper, requiring the balanced inclusion of (2)H_2O and H^+ on the left-hand side, and NH_4^+ on the right. Salts from viable reactions of specific acids were accepted.

(c) (i) Candidates who understood the $M^+ : [M+1]^+$ ratio were generally able to gain full credit here. Occasional arithmetical errors were seen.

(ii) Many candidates gave 'ester' as a correct answer but were then unable to corroborate this with sufficient and necessary information from the spectrum. Reference to an absorption outside of the C=O ester region contradicted an answer; equally, an absorption within the 1710–1750 cm^{-1} window was not solely sufficient, because of overlap with other absorptions. Corroboration could be given with reference to the C–O stretch (and the lack of O–H stretch).

C–H stretches are always ignored in these situations.

Absorptions must be accompanied by with assignment to an identified bond.

(iii) Propan-1-ol was a common incorrect answer for **T**. OH– is discouraged as a representation of an alcohol group. Many candidates did not draw the branched C5 acid correctly: some structures were unbranched; others had too many or too few H atoms.

3(a)(i)	$4P / P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**

3(c)(i)	H_2O / steam	**1**

3(c)(ii)		**1**

Question 3

(a) (i) Fully balanced equations were seen by more able candidates.

(ii) This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

(b) (i) Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

(ii) More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

(iii) Excellent answers were seen which identified phosphorus in H_3PO_3 as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*H_3PO_3 is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

(c) (i) Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

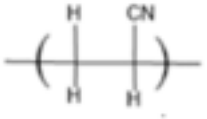
(ii) Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

(iii) Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

(iv) This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

(d) (i) More able candidates deduced a formula for the correct organic product in the equation. The inclusion of H_2O as the additional product was less common, even when included the equation was not always balanced.

(ii) The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

6(a)	 OR 	**1**
6(f)(i)	substitution	**1**
6(f)(ii)	**M1** ammonia / NH_3 **M2** heat + (M1 in ethanol) under pressure	**2**

Question 6

(a) The single repeat unit derived from the monomer, CH_2=CHCN, by addition polymerisation was quite well answered.

(b) (i) Many candidates recognised that the mechanism for step 1 was nucleophilic addition.

(ii) Overall, the mechanism for step 1 was generally not answered well. The polarity on the C=O bond was often absent and the arrow from the CN^- to the C (in the C=O bond) did not start from the lone pair on the C of the CN^- ion. Several answers contained an incorrect intermediate with a positive charge which was then attacked by the CN^- nucleophile and many answers incorrectly showed an arrow from an H^+ ion to an O^- (in the intermediate).

(iii) Some good answers were seen.

(iv) The absorptions in the infrared spectrum of CH_2=CHCN were often correctly identified as C≡N and C=C. Incorrect answers mainly contained an identification of the functional groups rather than the bond or identifying the absorption (annotated as T on the IR spectrum) as that of a C=O group.

(c) (i) Very few candidates gave an accurate description.

(ii) A number of answers were awarded some credit. Few candidates were able to give a full description.

(d) Several candidates gave a correct pair of molecules that were non-superimposable images of $CH_3CH(OH)CN$. A common error was diagrams containing the same type of bond at 180° to each other.

(e) The equation for the reduction of propanenitrile to propylamine using hydrogen gas was not well known. Few candidates used the correct stoichiometry for hydrogen.

(f) (i) It was well known that the name of the type of reaction in step 1 is substitution.

(ii) Many candidates gained credit for ammonia. Many omitted heat or the solvent as conditions.

3(a)(i)	KCN in ethanol + heat	**1**

Question 3

(a) (i) Some candidates incorrectly described the reagent and conditions required to convert carbonyl compounds to hydroxynitriles, rather than halogenoalkanes to nitriles.

(ii) The systematic name for **X** was not always used, e.g. propionitrile or ethyl cyanide. Omission of the 'e' in propanenitrile was often seen.

(iii) It was generally well known that this type of reaction is a hydrolysis reaction.

(iv) Many candidates correctly produced equations representing the production of hydrogen gas. The formula of the calcium salt was frequently given in terms of calcium with a valency of one.

(b) Many candidates knew that acidic conditions were required. Stronger candidates appreciated that the essential condition for this reaction was 'heat under reflux' to ensure complete oxidation to the carboxylic acid.

(c) Most candidates described the correct number of pi bonds present in **X**; the correct number of sigma bonds was seen less frequently. Seven, nine and ten sigma bonds were common incorrect answers.

3(a)	**M1** *appearance* (colourless / yellow) liquid **M2** *structure and bonding* simple / molecular **AND** covalent	**2**

3(d)(i)	ultraviolet light	**1**

Question 3

(a) Very few candidates could state the appearance of $SiCl_4$ at room temperature; more were able to state the correct structure and bonding. The most common incorrect response was giant covalent.

(b) Most candidates correctly referred to the appearance and pH. Candidates often did not refer to the relevant observation and struggled to differentiate the behaviours of the two chlorides.

(c) (i) A large number of candidates stated the correct reagent. Many left out aqueous or dilute as the conditions.

(ii) Many of the candidates defined disproportionation; a small proportion were able to refer to the correct oxidation numbers. Several referred incorrectly to the +1 oxidation number for chlorine.

(d) (i) This was well answered by most candidates.

(ii) Most candidates correctly answered the initiation step. Many candidates found the propagation and termination steps more challenging.

(e) This was well answered by most students.

(f) Many candidates were able to give the correct formulae at m/e = 164.Very few candidates gave both of the correct options at m/e = 166.

4(a)(i)	NaOH in alcohol / ethanol AND heat (under reflux)	**1**
4(b)(i)	cream(-coloured) / off-white precipitate (forms)	**1**
4(c)(i)	substitution	**1**

Question 4

This was a second organic chemistry question, which included some analytical work. This was generally answered well by many candidates, though later parts on IR and MS appeared less familiar to some.

(a) (i) 'Heat' or equivalent was often missing. H_2SO_4 was a popular answer despite the conditions suggested in the syllabus.

(ii) Many repeat units were shown containing four carbon atoms, as if the CH_3 groups had been incorporated into the backbone. Some candidates showed an ethene structure with missing CH_3 side groups.

(iii) Candidates very often showed the reverse equation with C_4H_8 and H_2O. Quite a few answers contained Al_2O_3 as a reactant and several answers omitted H_2O as a product.

(b) (i) This was well answered.

(ii) There were many correct structures/names for iodoform but very few correct propanoate ions.

(c) (i) There were many correct answers to this item.

(ii) Catalyst and oxidising agent were the common errors here, but otherwise there were many correct answers.

(d) (i) Compound **F** was commonly given incorrectly as the unknown compound, referring to an O–H absorption in the range 2850–3000 cm^{-1}. Candidates are reminded that C–H stretches appear in all organic spectra. Where absorptions are referenced, candidates need to identify which bond is stretching: 'nitrile', for example, is insufficient.

(ii) There were many null responses to this. Where attempted, candidates' answers were evenly split between correct identification of an identical abundance and some complicated (erroneous) mathematics.